

ME301 Lab 0 Report

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ABSTRACT

This introductory laboratory assignment explored the fundamental principles of sensing and actuation in robotics using an insect-like hexapod robot. A sonar sensor on the hexapod was utilized to measure distance readings, and multiple types of gaits were implemented through servo motor control. The sonar measurements demonstrated consistency and reliability by producing results with minimal variation. Three common hexapod gaits (wave, ripple, and tripod) were implemented with different step sizes and walk duration to test each gait's performance. The measured gait distances indicated tradeoffs between speed, directional accuracy, and stability across each gait type, with the tripod gait having the greatest forward displacement, the wave gait providing the greatest stability, and the ripple gait blending displacement and stability from the previous gaits. This lab highlights the importance of sensor processing and characterization, actuator coordination, and experimental strategy in robotic design.

I. INTRODUCTION

A robot is a mechanical autonomous system that exists in the physical world and is capable of sensing and interacting with its environment. Robots perceive their surroundings through sensors, process the retrieved information through computation, and produce corresponding actions using actuators and effectors. While robots vary widely, all robotic systems integrate sensors, actuators, effectors, and computation to enable interaction with their surroundings.

A sensor is a physical device that measures physical quantities such as temperature, light intensity, force, or distance [1, p. 70]. The type of quantity measured depends on the specific sensor being used. Sensors can also be categorized by the complexity of their required hardware and processing. Simple sensors typically require minimal processing power and hardware, while more complex sensors demand additional computation. Common examples of simple sensors include switches, bump switches, and light sensors. Basic switches provide one bit of information [1, p. 71], indicating a binary state such as on or off. Bump switches are a type of basic switch that detect physical contact with an object, outputting a signal when pressed. Light sensors, such as photoresistors, detect changes in light intensity by varying electrical resistance.

Sensor data is often noisy and must be processed to extract meaningful information. Raw sensor signals first undergo low-level electronic processing, where voltages or currents are measured. Signal processing techniques are then used to reduce noise and isolate relevant features of the data. Finally, high-level computation can be used to compare this signal with a larger database to more accurately interpret the data. The level of processing required depends on sensor complexity; for example, bump switches require minimal processing, while light or distance sensors often require filtering to improve reliability.

Once sensor data is processed, robots produce motion through actuators and effectors. Effectors are the components that directly interact with the environment, such as legs, wheels, or grippers [1, p. 29]. Actuators are the mechanisms that drive these effectors. Common actuator types include electric motors, hydraulic cylinders, pneumatic systems, and smart materials. Electric motors are the most widely used actuators in robotics. DC motors are a type of electric motor that generate continuous rotational motion by applying a voltage across the motor, making them suitable for tasks requiring variable speed. Servo motors, a specialized form of DC motors with integrated gears, sensors, and controllers, enable precise position control. By sending signals that specify a desired angle, servo motors allow robots to achieve accurate actuation. The coordination of multiple actuators allow robots to perform complex tasks such as object manipulation.

This lab assignment focuses on developing familiarity with robotic actuation and sensing through hands-on interaction and software development of a hexapod robot. By identifying servo motor mappings, characterizing sonar measurements, and implementing common gait patterns, the lab establishes foundational skills in robot control, data collection, and experimental strategy that will be built upon in future robotics lab assignments.

II. METHODS

The robot used in this lab assignment is a hexapod robot equipped with six legs, each driven by servo motors that enable controlled joint motion. The hexapod robot platform used in this lab is shown in Figure 1.



Figure 1. Hexapod Robot

Each leg of the hexapod contains three servo motors, allowing for complex leg articulation. The servo closest to the body enables forward and backward leg motion. The second servo, located in the middle of the leg, controls vertical leg movement, allowing the leg to lift off and return to the ground. The final servo controls the knee joint, enabling inward and outward bending of the leg. Each servo receives position commands from the robot's microcontroller. By coordinating precise servo positions across all joints, smooth and complex locomotion can be achieved.

In addition to its actuators, the hexapod is equipped with a forward-facing sonar sensor used to measure the distance between the robot and nearby objects. The sonar sensor, shown in Figure 2, operates by emitting an ultrasonic pulse and measuring the time delay until the reflected signal returns to the sensor. This delay measurement is converted into distance using the known speed of sound.

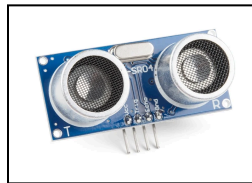


Figure 2. Sonar Sensor on Hexapod Robot

A `readSonar()` function was implemented to read and display the sonar sensor output for a duration of one second. When the function is called, an `init_time` variable is recorded using the `time.time()` function from the Python time library. A while loop continuously outputs sonar readings until the elapsed time exceeds one second. Distance measurements were obtained by calling the `getDistance()` function from the sonar library. To verify correct functionality, a moving object was placed in front of the sensor, and changes in the reported distance were observed.

To characterize the sonar sensor, distance measurements were collected at three known distances: 10 cm, 20 cm, and 30 cm. A `readMultipleSonar()` function was developed to automate data collection by prompting the user to

position an object at a specified distance. Once confirmed, the sonar sensor recorded 10 distance measurements. These values were used to calculate the mean distance and standard deviation for each test position.

After characterizing the sonar sensor, three walking gaits were implemented using the robot SDK: wave, ripple, and tripod. Each gait was implemented with both a small step size and a larger step size to evaluate the effect of motor command amplitude on walking distance.

The wave gait is characterized by moving one leg at a time while the remaining legs maintain ground contact. This gait was implemented by lifting a single leg using the vertical servo while simultaneously swinging the leg forward using the forward/backward servo. The leg was then lowered back to the ground before the next leg repeated the motion. After all legs completed this sequence, the forward/backward servos returned to their home positions, effectively pulling the robot body forward.

The ripple gait is similar to the wave gait, but two legs move simultaneously instead of one. The six legs were grouped into three pairs, as shown in Table 1. Additionally, Table 1 shows the groups for a tripod gait. A tripod gait moves three legs simultaneously compared to a ripple's two legs and a wave's one leg.

Table 1. Ripple and Tripod Groups

Group #	Ripple Gait	Tripod Gait
1	Front Left Leg, Middle Right Leg	Front Right Leg, Middle Left Leg, Bottom Right Leg
2	Middle Left Leg, Bottom Right Leg	Front Left Leg, Middle Right Leg, Bottom Left Leg
3	Bottom Left Leg, Top Right Leg	N/A

During each ripple/tripod iteration, one group of legs lifted and moved forward while the remaining groups remained grounded and actuated backward. This coordinated motion resulted in forward translation of the robot body.

Gait performance was evaluated by measuring the distance traveled over fixed time intervals. Displacement in both the x and y directions was recorded for each gait over durations of 10 seconds and 30 seconds, using both small and large step sizes. Comparing these measurements allowed for direct evaluation of gait efficiency and speed.

In the code, each of the functions (sonar, gait types) were implemented such that a prompt would ask the user which function they would like to call. The sonar functions used user input to test which specific distance the user was testing. The gait types utilized user input to determine gait type, step size, and gait duration. By asking for user input, this method allowed for efficient testing of all functions.

III. RESULTS

The sonar readings from the 10cm, 20cm, and 30cm are listed in millimeters below in Table 2.

Table 2. Sonar Sensor Distance Readings from 10cm, 20cm, 30cm

Test #	10cm Away (mm)	20cm Away (mm)	30cm Away (mm)
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1	130	230	340
2	130	230	340
3	130	230	340
4	130	230	340
5	130	230	340
6	130	230	340
7	131	230	340
8	130	230	340
9	131	230	339
10	130	230	340

This data was collected by reading the output of the readMultipleSonar() function after placing an object at 10cm, 20cm, and 30cm in front of the sensor. The same object was used for each of the trials to maintain consistency. The means and standard deviations of each distance is reported in Table 3.

Table 3. Mean and Standard Deviation of Sonar Readings at 10cm, 20cm, 30cm

	10cm (mm)	20cm (mm)	30cm (mm)
Mean	130.1	230	339.3
Std. Dev.	0.3	0	0.3

The distance traveled by each of the gaits are recorded below in Table 4. Each gait for a time duration and step size was tested 3 times.

Table 4. Distance Traveled for Each Gait with Small/Large Step and 10/30 sec. with Mean and Standard Deviation

(cm)	Wave, Small	Wave, Large	Tripod, Small	Tripod, Large	Ripple, Small	Ripple, Large
Distance traveled (x,y) in 10 seconds	(14,2) (14,2) (14,2)	(26,9) (26,9) (26,9)	(69.6, 1.5) (69.4, 2.6) (69.2, 4.4)	(153, 5.3) (153, 4) (151,1)	(22.5,5.5) (22.5,5.5) (22.5,5.5)	(36,8.5) (36,8.5) (36,8.5)
Mean	(14, 2)	(26, 9)	(69.4, 2.83)	(152.3, 3.43)	(22.5, 5.5)	(36, 8.5)
Std. Dev.	(0, 0)	(0, 0)	(0.4, 1.09)	(0.93, 1.8)	(0, 0)	(0, 0)
Distance traveled (x,y) in 30 seconds	(29.5,-6.75) (32,4) (32,6.75)	(58,19.5) (56,26.5) (59.5,20)	(183,29) (179,26) (196.5,41)	(399.5,53.25) (391,78) (386,106)	(74, 0) (74.5, 12) (75, 11.25)	(119.5,25.5) (117, 5) (117,19.75)

Mean	(31.16, 1.33)	(57.66, 22)	(186, 32)	(392.16, 79.08)	(74.33, 7.75)	(117.83, 16.75)
Std. Dev.	(1.17, 5.82)	(1.24, 3.18)	(7.25, 6.48)	(5.57, 21.55)	(0.47, 5.48)	(1.18, 8.63)

The distance traveled by the hexapod for each trial was recorded by measuring the distance of an origin to the final position with a ruler after the gait completed its motion. The origin of the hexapod was set as the center of the sonar sensor. The forward-facing direction of the sonar sensor was designated as the positive x-direction, while the positive y-direction was defined as pointing toward the left side of the hexapod.

All gait testing was conducted on a dusty, flat surface. At the beginning of each trial, the hexapod was placed at the same starting location and orientation to ensure consistency across measurements.

IV. DISCUSSION

The sonar sensor displayed consistent results for all the tested distances, with minimal variations per trial. The small standard deviations at each distance indicate that the recorded values were tightly clustered, confirming that the sensor provides reliable measurements under controlled, static conditions. This consistency suggests that the sonar sensor is well-suited for basic distance sensing tasks.

All recorded sonar outputs were in the hundreds, indicating that the sensor reports distances in millimeters. However, the measured distances were consistently approximately 30 mm greater than the intended object distances. This systematic offset is likely due to a misinterpretation of the sensor's physical reference point. During testing, the zero-distance reference was assumed to be at the front of the sonar sensor rings. Measuring the physical length of the sonar ring revealed a length of approximately 20 mm, accounting for a majority of the observed offset. The remaining discrepancy of roughly 10 mm may be attributed to factors such as sensor placement error, signal processing delays, or internal calibration inaccuracies.

Despite this offset, the sonar sensor maintained a strong linear relationship between measured distance and actual distance. This linearity is critical in robotic perception, as systematic offsets can be compensated for through calibration, whereas large random variations would significantly reduce measurement reliability. In contrast, a sensor with a large standard deviation would be difficult to interpret due to the presence of substantial noise, even if its average readings were accurate.

The consistency observed under static testing conditions highlights the importance of controlled environments when characterizing sensors. However, a key limitation of this experiment is that measurements were obtained using a single object with a fixed orientation and material. In real-world applications, objects vary in shape, surface properties, and orientation, all of which can affect sonar reflection quality. This limitation highlights the challenge of relying on a single sensor and motivates the use of sensor fusion in more advanced robotic systems to improve robustness and reliability.

The gait experiments revealed clear performance differences among the implemented strategies. The tripod gait consistently produced the largest forward displacements compared to the wave and ripple gaits. This outcome aligned with expectations, as the tripod gait allows three legs to simultaneously provide propulsion during each gait cycle, resulting in greater forward force. However, the tripod gait also exhibited the largest lateral displacement. One possible explanation is that having more legs pushing against the ground increases interaction with the dusty surface, leading to uneven friction forces and increased sideways drift.

In contrast, the wave gait produced significantly smaller forward displacements. Because only one leg moves at a time, each full gait cycle requires all six legs to complete their motion before the robot advances. This results in slower overall movement. However, the wave gait exhibited the smallest standard deviation and minimal lateral movement, indicating greater stability and directional consistency.

The ripple gait demonstrated performance characteristics intermediate between the wave and tripod gaits. By moving two legs at a time, the ripple gait achieved faster forward motion than the wave gait while maintaining greater stability than the tripod gait. Its moderate standard deviation reflects this balance between speed and stability. In general, larger step sizes and faster motor speeds resulted in increased lateral displacement and greater variability, highlighting the tradeoff between speed and control.

Although the measurements were reasonably accurate, there were several sources of uncertainty. Because the robot was manually repositioned at the start of each trial, small variations in initial placement were unavoidable. Additionally, final positions were marked using tape, making it difficult to identify an exact endpoint, which may have introduced small measurement offsets. Other sources of noise include servo motor jitter, particularly when the battery voltage dropped to approximately 8.5V. At lower voltages, the motors exhibited inconsistent behavior, which likely contributed to variability in gait performance.

V. CONCLUSION

This lab provided hands-on experience and testing with fundamental components of robotics including sensing and actuation. With a sonar sensor and multiple common hexapod gaits, the experiments and collected data demonstrated how to interpret sensor data and strategically approach gait implementation.

The sonar sensor showed high consistency and reliability under controlled, simple environmental conditions. The sensor demonstrated a linear pattern between the sensor output and the actual tested distance. However, there was a prevalent measured offset due to a combination of center placement and signal processing errors. The consistent sensor and the offset being due to mainly human error showed that the offset could be compensated through post-data collection interpretation or sensor calibration. The sensor testing highlighted the importance of understanding the capabilities and limitations of sensors before implementing them into robot control or data collection.

The gait experiments highlighted clear strengths and weaknesses in each of the tested gaits. The tripod demonstrated large forward movement, but showed equally significant lateral displacement. On the other hand, the wave gait maintained stability and directional consistency at the cost of speed. The ripple gait demonstrated a blend of both gait, balancing speed and stability. Interpreting each of the gait displacements showed that an implementation of a specific gait type is highly dependent on the type of environmental conditions and ultimate goal of the robot.

Overall, this introductory lab highlighted the considerations, tradeoffs, and challenges that play a role in robotic design. The traversed environment and intended application determine which sensors, actuators, and effectors are needed to properly accomplish the goal. Outside noise, testing conditions, and human error can affect the performance and consistency of a robot. The understanding and considerations of the robot's mechanisms and limitations help establish foundational skills to build more advanced robotic systems in the future.

REFERENCES

- [1] “The robotics primer,” *Choice Reviews Online*, vol. 45, no. 06, pp. 45–322245–3222, Feb. 2008, doi: <https://doi.org/10.5860/choice.45-3222>.